

$$\text{or, } \ln x = \ln(y) = v + \ln c \quad y/x + \ln e$$

$$\text{or, } \ln x = \frac{y}{x} - \ln \frac{y}{x} + \ln c = \frac{y}{x} + \ln \frac{x}{y} + \ln c$$

$$\text{or, } \ln \frac{x \cdot y}{x \cdot c} = \frac{y}{x} \quad \text{or,}$$

$$\text{or, } \ln x + \ln y = v + \ln c \quad \text{or, } \ln x + \ln y - \ln c = v$$

$$\text{or, } \ln \frac{xy}{c} = v \quad \text{or, } \frac{xy}{c} = e^v \quad \text{or, } \frac{xy}{c} = c e^v$$

$$\text{or, } \frac{xy}{c} = c e^{y/x} \quad \text{or, } y = c e^{y/x}$$

which is required solution.

Equation Reducible to Homogeneous Form

Working Rules: If any equation of the form

$$\frac{dy}{dx} = \frac{ax+by+c}{a_1x+b_1y+c_1} \quad \text{--- (1)}$$

where $\frac{a}{a_1} \neq \frac{b}{b_1}$ then put $x = X+h$ & $y = Y+k$

$$\text{so that } \frac{dy}{dx} = \frac{dY}{dX}$$

where X & Y are new variable and h & k are arbitrary constant.

so eqn (1) becomes,

$$\frac{dY}{dX} = \frac{aX+bY+(ah+bk+c)}{a_1X+b_1Y+(a_1h+b_1k+c_1)} \quad \text{--- (2)}$$

choosing the values of h & k such that,

$$ah + bk + c = 0 \quad \& \quad a_1h + b_1k + c_1 = 0$$

এইটো cross multiplication method দিচ্ছি

solve করে (2) no eqn এ করে

$$\therefore \frac{dy}{dx} = \frac{ax + by}{a_1x + b_1y} \text{ which is homogeneous}$$

diff. eqn.

যাকিছুকিছু homogeneous diff. এর নিয়মে,

Ex-3 | Solve $(2x + y + 3) \frac{dy}{dx} = x + 2y + 3$

Soln:

Given that,

$$(2x + y + 3) \frac{dy}{dx} = x + 2y + 3$$

$$\text{or, } \frac{dy}{dx} = \frac{x + 2y + 3}{2x + y + 3} \quad \text{--- (1)} \quad \frac{\frac{1}{2}}{\frac{1}{2}} \neq \frac{\frac{2}{2}}{\frac{1}{2}}$$

Put $x = X + h$ and $y = Y + k$ so that $\frac{dy}{dx} = \frac{dY}{dX}$

so eqn (1) becomes,

$$\frac{dY}{dX} = \frac{X + 2Y + (h + 2k + 3)}{2X + Y + (2h + k + 3)} \quad \text{--- (2)}$$

choosing the values of h and k such that,

$$h + 2k + 3 = 0 \quad \text{--- (3)}$$

$$\text{and } 2h + k + 3 = 0 \quad \text{--- (4)}$$

Solving (3) and (4) we get,

$$\frac{h}{6-3} = \frac{k}{6-3} = \frac{1}{1-4}$$

$$\therefore h = -1 \text{ and } k = -1$$

Now from eqn (2),

$$\frac{dY}{dX} = \frac{X + 2Y}{2X + Y} \quad \text{--- (5)}$$

which is homogeneous eqn.

Now putting $Y = vX$

$$\text{so that } \frac{dY}{dX} = v + X \frac{dv}{dX}$$

so eqn (5) becomes,

$$v + X \frac{dv}{dX} = \frac{X + 2vX}{2X + vX}$$

$$\begin{aligned} \text{or, } X \frac{dv}{dX} &= \frac{X + 2vX}{2X + vX} - v = \frac{X + 2vX - 2vX - v^2X}{2X + vX} \\ &= \frac{1 - v^2}{2 + v} \end{aligned}$$

$$\text{or, } \frac{dX}{X} = \left(\frac{2+v}{1-v^2} \right) dv$$

$$\Rightarrow 0\pi, \frac{dx}{x} = \left(\frac{3/2}{1-u} + \frac{1/2}{1+u} \right) du$$

Now Integrating,

$$\ln x = -\frac{3}{2} \ln(1-u) + \frac{1}{2} \ln(1+u)$$

$$0\pi, \ln x = \ln \frac{(1+u)^{1/2}}{(1-u)^{3/2}}$$

$$0\pi, x = \frac{(1+u)^{1/2}}{(1-u)^{3/2}}$$

$$0\pi, x^2 = \frac{(1+u)}{(1-u)^3}$$

$$= \frac{\left(1 + \frac{y}{x}\right)}{\left(1 - \frac{y}{x}\right)^3}$$

$$0\pi, \left(1 - \frac{y}{x}\right)^3 x^2 = 1 + \frac{y}{x}$$

$$0\pi, \left(\frac{x-y}{x}\right)^3 x^2 = \frac{x+y}{x}$$

$$0\pi, (x-y)^3 x = x+y$$

$$0\pi, (x-y)^3 x = x+y+1$$

$$0\pi, (x-y)^3 x = x+y+2$$

which is required solution.

$$\frac{1}{1+u} \quad \frac{1}{1-u}$$

$$1+u=0 \therefore u=-1$$

$$\therefore 1-(-1)=2$$

$$\therefore \frac{2+(-1)}{2} = \frac{1}{2}$$

$$1-u=0 \therefore u=1$$

$$\therefore \frac{2+1}{1+(1)} = \frac{3}{2}$$

Ex-1 solve $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ $\frac{a}{a_1} \neq \frac{b}{b_1}$

Solⁿ: Given that,

$$\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3} \quad \text{--- (1)}$$

Put $x = X+h$ and $y = Y+k$ so that $\frac{dy}{dx} = \frac{dY}{dX}$

so eqn (1) becomes

$$\frac{dY}{dX} = \frac{X+2Y+(h+2k-3)}{2X+Y+(2h+k-3)} \quad \text{--- (2)}$$

choosing the value of h and k such that,

$$h+2k-3=0 \quad \text{--- (3)}$$

$$2h+k-3=0 \quad \text{--- (4)}$$

solving eqn (3) and (4) we get,

$$\frac{h}{-6+3} = \frac{k}{-6+3} = \frac{1}{1-4}$$

$$\therefore h = 1 \quad \text{and} \quad k = 1$$

Now from eqn (2),

$$\frac{dY}{dX} = \frac{X+2Y}{2X+Y} \quad \text{--- (5)}$$

Now putting $Y=vX$ so that $\frac{dY}{dX} = v + X \frac{dv}{dX}$

So eqn (5) becomes,

$$\cancel{dx} (v + X \frac{dv}{dX}) = \frac{X + 2vX}{2X + vX}$$

$$\text{or, } X \frac{dv}{dX} = \frac{1+2v}{2+v} - v$$

$$\text{or, } X \frac{dv}{dX} = \frac{1+2v-2v-v^2}{2+v}$$

$$= \frac{1-v^2}{2+v}$$

$$\frac{dX}{X} = \left(\frac{2+v}{1-v^2} \right) dv$$

$$= \left\{ \frac{-\frac{1}{2}(-2v) + 2}{1-v^2} \right\} dv$$

$$= \left\{ \frac{-\frac{1}{2}(-2v)}{1-v^2} + \frac{2}{1-v^2} \right\} dv$$

Now Integrating,

$$\ln X = -\frac{1}{2} \ln(1-v^2) + 2 \cdot \frac{1}{2 \cdot 1} \ln \frac{1+v}{1-v} \quad \left| \begin{array}{l} \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \\ \ln\left(\frac{a+x}{a-x}\right) \end{array} \right.$$

$$= \ln \frac{(1+v)}{(1-v)(1-v^2)^{1/2}} + \ln C$$

$$\ln x = \ln \frac{(1+u)^{1/2}}{(1-u)^{3/2}} + \ln C$$

$$\text{or, } x = C \frac{(1+u)^{1/2}}{(1-u)^{3/2}}$$

$$\text{or, } x^2 = C^2 \frac{1+u}{(1-u)^3} = C^2 \frac{1 + \frac{y}{x}}{\left(1 - \frac{y}{x}\right)^3}$$

$$\text{or, } x^2 = C^2 \frac{\frac{x+y}{x}}{\left(\frac{x-y}{x}\right)^3}$$

$$\text{or, } 1 = C^2 \frac{x-1+y-1}{(x-1-y+1)^3} = C^2 \frac{x+y-2}{(x-y)^3}$$

$\therefore (x-y)^3 = C^2(x+y-2)$ is the required solution.

Ex-2 | solve $\frac{dy}{dx} = \frac{3y-7x+7}{3x-7y-3}$ $\frac{a}{a_1} = \frac{-7}{3}, \frac{b}{b_1} = \frac{3}{-7}$

Soln: Given that,

$$\frac{dy}{dx} = \frac{3y-7x+7}{3x-7y-3} \quad \text{--- (1)}$$

Put $x = X+h$ and $y = Y+k$ so that $\frac{dy}{dx} = \frac{dY}{dX}$

So eqn (1) becomes,

$$\frac{dY}{dX} = \frac{-7X+3Y+(-7h+3k+7)}{3X-7Y+(3h-7k-3)} \quad \text{--- (2)}$$

Choosing the value of h and k such that,

$$-7h + 3k + 7 = 0 \quad \text{--- (3)}$$

$$3h - 7k - 3 = 0 \quad \text{--- (4)}$$

Solving eqn (3) and (4) we get,

$$\frac{h}{-9 + 49} = \frac{k}{21 - 21} = \frac{1}{49 - 9}$$

$$\therefore h = 1 \text{ and } k = 0$$

Now from eqn (2),

$$\frac{dY}{dX} = \frac{-7X + 3Y}{3X - 7Y} \quad \text{--- (5)}$$

Now putting $Y = vX$ so that $\frac{dY}{dX} = v + X \frac{dv}{dX}$

So eqn (5) becomes,

$$v + X \frac{dv}{dX} = \frac{-7X + 3vX}{3X - 7vX} = \frac{-7 + 3v}{3 - 7v}$$

$$\text{or, } X \frac{dv}{dX} = \frac{-7 + 3v}{3 - 7v} - v = \frac{-7 + 3v - 3v + 7v^2}{3 - 7v}$$

$$= \frac{-7 + 7v^2}{3 - 7v}$$

$$\text{or, } 7 \frac{dX}{X} = \left(\frac{3 - 7v}{v^2 - 1} \right) dv$$

$$7 \frac{dx}{x} = - \left(\frac{2}{v-1} + \frac{5}{v+1} \right) dv$$

Now Integrating,

$$7 \ln x = -2 \ln(v-1) - 5 \ln(v+1) + \ln c$$

~~$$\neq \ln c (v-1)^2$$~~

$$\text{or, } \ln x^7 = \ln \frac{c}{(v-1)^2 (v+1)^5}$$

$$\text{or, } x^7 (v-1)^2 (v+1)^5 = c$$

$$\text{or, } x^7 \left(\frac{y}{x} - 1 \right)^2 \left(\frac{y}{x} + 1 \right)^5 = c$$

$$\text{or, } x^7 \left(\frac{y-x}{x} \right)^2 \left(\frac{y+x}{x} \right)^5 = c$$

$$\text{or, } (y-x)^2 (y+x)^5 = c \quad \text{or, } (y-x+1)^2 (y+x-1)^5 = c$$

which is the required solution.

Ex-4 Solve $(2x-2y+5) \frac{dy}{dx} = x-y+3$

Soln: Given that,

$$(2x-2y+5) \frac{dy}{dx} = x-y+3$$

$$\text{or, } \frac{dy}{dx} = \frac{x-y+3}{2x-2y+5}$$

$$\frac{3-7v}{v+1} = \frac{3-7(-1)}{-1-1}$$

$$v+1=0 \therefore v=-1$$

$$\frac{3-7v}{v-1} = \frac{3-7(-1)}{-1-1}$$

$$= \frac{10}{-2} = -5$$

$$v-1=0 \therefore v=1$$

$$\frac{3-7v}{v+1} = \frac{3-7}{1+1} = \frac{-4}{2} = -2$$

Special case of Homogeneous differential eqn:

Working rule: If any eqn of the form

$$\frac{dy}{dx} = \frac{ax+by+c}{a_1x+b_1y+c_1} \quad (1) \text{ where } \frac{a}{a_1} = \frac{b}{b_1}$$

then let $ax+by = v$ [$ax+by$ term]

so that,

$$a + b \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{1}{b} \left(\frac{dv}{dx} - a \right)$$

Putting this in given eqn then separating variable and integral

Ex-4 soln:

$$\frac{dy}{dx} = \frac{x-y+3}{2(x-y)+5} \quad (1)$$

Let, $x-y = v$

$$\therefore 1 - \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \frac{dy}{dx} = 1 - \frac{dv}{dx}$$

So eqn (1) become,

$$1 - \frac{dv}{dx} = \frac{v+3}{2v+5}$$

$$\text{or, } \frac{dv}{dx} = 1 - \frac{v+3}{2v+5} = \frac{2v+5-v-3}{2v+5} = \frac{v+2}{2v+5}$$

$$\text{or, } dx = \left(\frac{2v+5}{v+2} \right) dv = \frac{2(v+2)+1}{v+2} dv$$

$$= \left(\frac{2}{v+2} + \frac{1}{v+2} \right) dv$$

$$\text{Now Integrating, } x+c = 2 \ln(v+2) + \ln(v+2)$$

$$= \ln(v+2)^3$$

$$= \ln(x-y+2)^3$$

$$= 2(x-y) + \ln(x-y+2)$$

$$\therefore 2y - x + c = \ln(x-y+2) ; \text{ which is}$$

required solution.

Ex-5 | solve $\frac{dy}{dx} = \frac{6x-4y+3}{3x-2y+1}$ | $\frac{a}{a_1} = \frac{6}{3} = 2, \frac{b}{b_1} = \frac{-4}{-2} = 2$

Soln: Given that,

$$\frac{dy}{dx} = \frac{6x-4y+3}{3x-2y+1}$$

$$\text{or, } \frac{dy}{dx} = \frac{2(3x-2y)+3}{3x-2y+1} \quad \text{--- (1)}$$

$$\text{Let, } 3x - 2y = v$$

$$\text{or, } 3 - 2 \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{3 - \frac{dv}{dx}}{2}$$

So eqn (1) become,

$$\frac{3 - \frac{dv}{dx}}{2} = \frac{2v + 3}{v + 1}$$

$$\text{or, } 3 - \frac{dv}{dx} = \frac{4v + 6}{v + 1}$$

$$\text{or, } \frac{dv}{dx} = 3 - \frac{4v + 6}{v + 1} = \frac{3v + 3 - 4v - 6}{v + 1}$$

$$= \frac{-(v + 3)}{v + 1}$$

$$\text{or, } dx = \frac{v + 1}{-(v + 3)} dv$$

$$= - \frac{v + 3 - 2}{v + 3} dv$$

$$= - \left(1 - \frac{2}{v + 3} \right) dv$$

Now Integrating,

$$x = - \left\{ v - 2 \ln(v + 3) \right\} + c$$

$$= -v + 2 \ln(v + 3) + c$$

$$\text{on, } x = 2y - 3x + 2\ln(3x - 2y + 3) + c$$

$$\text{on, } 4x - 2y = 2\ln(3x - 2y + 3) + c$$

$\therefore 2x - y = \ln(3x - 2y + 3) + c/2$ which is the required solution

Ex-6 | Solve $(6x - 4y + 1) \frac{dy}{dx} = 3x - 2y + 1$

Soln : Given that,

$$\frac{dy}{dx} = \frac{3x - 2y + 1}{6x - 4y + 1}$$

$$\left| \begin{array}{l} \frac{a}{a_1} = \frac{1}{2} \\ \frac{b}{b_1} = \frac{1}{2} \end{array} \right.$$

$$= \frac{(3x - 2y) + 1}{2(3x - 2y) + 1} \quad \text{--- (1)}$$

Let, $3x - 2y = u \quad \therefore 3 - 2 \frac{dy}{dx} = \frac{du}{dx}$

$$\therefore \frac{dy}{dx} = \frac{3 - \frac{du}{dx}}{2}$$

So eqn (1) become,

$$\frac{3 - \frac{du}{dx}}{2} = \frac{u + 1}{2u + 1}$$

$$\text{on, } \frac{du}{dx} = 3 - \frac{2u + 2}{2u + 1} = \frac{6u + 3 - 2u - 2}{2u + 1}$$

$$\frac{dv}{dx} = \frac{4v+1}{2v+1}$$

$$\text{or, } dx = \frac{2v+1}{4v+1} dv = \frac{(4v+1) \cdot \frac{1}{2} + \frac{1}{2}}{4v+1} dv$$

$$= \int \left(\frac{1}{2} + \frac{1}{2(4v+1)} \right) dv$$

$$\text{or, } \cancel{2x} = 2x = \frac{1}{2} v + \frac{1}{8} \ln(4v+1)$$

Now Integrating,

$$\text{or, } 2x = 3x - 2y + \ln(4(3x - 2y) + 1) + C$$

$$\text{or, } 2y - x = \ln(12x - 8y + 1) + C$$

which is the required solution.

Linear Differential Eqn :

Definition: A differential equation of the form $\frac{dy}{dx} + Py = Q$; where P & Q are the function of x or constant is called linear differential eqn.

Working rule: If any differential eqn of the form $\frac{dy}{dx} + Py = Q$ the I.F (Integrating Factor) $= e^{\int P dx}$ which multiplying $\leftrightarrow$ given eqn we get,

$$e^{\int P dx} \cdot \frac{dy}{dx} + Py e^{\int P dx} = Q e^{\int P dx}$$

$$\text{or, } \frac{d}{dx} (e^{\int P dx} \cdot y) = Q \cdot e^{\int P dx}$$

[always $\frac{d}{dx}(uv)$ এর সূত্র
এর আকার নিয়ে]

$$\text{Integrating, } e^{\int P dx} y = \int Q e^{\int P dx} + c$$

which is required solution.

^{Ex-1}
~~Q:~~ Solve $(1-x^2) \frac{dy}{dx} - xy = 1$

এখানে y dependent

Soln: Given that,

$$(1-x^2) \frac{dy}{dx} - xy = 1$$

$$\text{or, } \frac{dy}{dx} + \left(\frac{-x}{1-x^2} \right) y = \frac{1}{1-x^2} \quad \text{--- (1)}$$

Now,

$$I.F. = e^{\int -\frac{x}{1-x^2} dx}$$

$$= e^{\frac{1}{2} \int -\frac{2x}{1-x^2} dx}$$

$$= e^{\frac{1}{2} \ln(1-x^2)} = e^{\ln(1-x^2)^{1/2}}$$

$$= \sqrt{1-x^2}$$

Now multiplying eqn (1) by $(\sqrt{1-x^2})$ we get,

$$\sqrt{1-x^2} \frac{dy}{dx} - \frac{x}{1-x^2} (\sqrt{1-x^2}) y = \frac{1}{1-x^2} \sqrt{1-x^2}$$

$$\text{or, } \frac{d}{dx} (\sqrt{1-x^2} \cdot y) = \frac{1}{\sqrt{1-x^2}}$$

Now Integrating we get,

$$\sqrt{1-x^2} \cdot y = \sin^{-1} x + C$$

which is required solution.

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x$$

Ex² Q: Solve $x \frac{dy}{dx} + 2y = x^2 \log x$

Soln: Given that,

$$x \frac{dy}{dx} + 2y = x^2 \log x$$

$$\text{or, } \frac{dy}{dx} + \frac{2}{x} y = x \log x \quad \text{--- (1)}$$

y dependent variable

Now I.F. = $e^{\int \frac{2}{x} dx} = e^{\cancel{2}} e^{2 \ln x} = x^2$

Now multiplying eqn (1) by x^2 we get,

$$x^2 \frac{dy}{dx} + 2xy = x^3 \log x \quad \text{or, } \frac{d}{dx}(x^2 y) = x^3 \log x$$

Now Integrating, $x^2 y = \int x^3 \log x dx + c$

$\int u v dx = u \int v dx -$

$\int \left\{ \frac{du}{dx} \int v dx \right\} dx$

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अक्ष (म व.)

$$\begin{aligned} \text{or, } x^2 y &= \log x \int x^3 dx - \int \left\{ \frac{d}{dx} (\log x) \int x^3 dx \right\} dx + c \\ &= \log x \cdot \frac{x^4}{4} - \int \left(\frac{1}{x} \cdot \frac{x^4}{4} \right) dx + c \\ x^2 y &= \log x \cdot \frac{x^4}{4} - \frac{x^4}{16} + c \end{aligned}$$

$$\therefore y = Cx^{-2} + \frac{1}{4} x^2 (\log x - \frac{1}{4})$$

which is the required solution. (

Ex: 2(b) solve $x \frac{dy}{dx} + 2y = x^4$ | y dependent

Soln: Given that,

$$x \frac{dy}{dx} + 2y = x^4$$

$$\text{or, } \frac{dy}{dx} + \frac{2}{x} y = x^3 \quad \text{--- (1)}$$

Now I.F. = $e^{\int \frac{2}{x} dx} = x^2$

Now multiplying eqn (1) by x^2 we get,

$$x^2 \frac{dy}{dx} + 2xy = x^5$$

$$\text{or, } \frac{d}{dx}(x^2 y) = x = x^5$$

Now integrating, $x^2 y = \frac{x^6}{6} + c$

which is required solⁿ.

Ex-3 Solve $(x^3 - x) \frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x$
 $| y \text{ dependent}$

Solⁿ: Given, $(x^3 - x) \frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x$

$$\text{or, } \frac{dy}{dx} - \frac{3x^2 - 1}{x^3 - x} y = \frac{x^5 - 2x^3 + x}{x^3 - x}$$

$$= \frac{x^5 - x^3 - x^3 + x}{x(x^2 - 1)}$$

$$= \frac{x^3(x^2 - 1) - x(x^2 - 1)}{x(x^2 - 1)}$$

$$= \frac{(x^3 - x)(x^2 - 1)}{x(x^2 - 1)}$$

$$\text{or, } \frac{dy}{dx} - \frac{3x^2 - 1}{x^3 - x} y = x^2 - 1 \quad (1)$$

Now, I.F. = $e^{\int (-\frac{3x^2 - 1}{x^3 - x}) dx}$

$$= e^{-\ln(x^3 - x)}$$

$$= \frac{1}{x^3 - x}$$

Now multiplying eqn(1) by $\frac{1}{x^3-x}$ we get,

$$\frac{1}{x^3-x} \frac{dy}{dx} - \frac{3x^2-1}{(x^3-x)^2} y = \frac{x^2-1}{x^3-x}$$

$$\text{or, } \frac{d}{dx} \left(\frac{y}{x^3-x} \right) = 0 + \int \frac{x^2-1}{x^3-x} dx = 0 + \int \frac{3x^2-1}{x^3-x} dx$$

$$\text{Now Integrating, } = 0 + \int \frac{\frac{1}{3}(3x^2-1)}{x^3-x} dx$$

$$\frac{y}{x^3-x} = 0 + \int \frac{dx}{x} = 0 + \ln x$$

which is the required solution.

Ex-8 Solve $x \frac{dy}{dx} + 2y = \frac{dy}{dx} + 4$ | y dependent

Soln: Given that,

$$x \frac{dy}{dx} + 2y = \frac{dy}{dx} + 4$$

$$\text{or, } (x-1) \frac{dy}{dx} + 2y = 4$$

$$\text{or, } \frac{dy}{dx} + \frac{2}{x-1} y = \frac{4}{x-1} \quad \text{--- (1)}$$

$$\text{Now I.F.} = e^{\int \frac{2}{x-1} dx} = e^{2 \ln(x-1)} = (x-1)^2$$

Now multiplying eqn (1) by $(x-1)^2$ we get,

$$(x-1)^2 \frac{dy}{dx} + 2(x-1)y = 4(x-1)$$

$$\text{or, } \frac{d}{dx} ((x-1)^2 y) = 4(x-1)$$

Now Integrating,

$$(x-1)^2 y = c + 4 \int (x-1) dx$$

$$\text{or, } y(x-1)^2 = c + 4 \left(\frac{x^2}{2} - x \right)$$

$$= c + 2(x^2 - 2x)$$

which is the required solution

Equation which become linear when x is treated as dependent variable.

$$\text{Form : } \frac{dx}{dy} + Px = Q$$

$$\text{I.F.} = e^{\int P dy}$$

Ex-1 Solve $y \log y dx + (x - \log y) dy = 0$

এখানে y বসিয়ে x কাম করে x dependent variable.

Soln: Given that,

$$y \log y dx + (x - \log y) dy = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{x - \log y}{y \log y}$$

$$\text{or, } y \log y dx = -(x - \log y) dy$$

$$\text{or, } \frac{dx}{dy} + \frac{x - \log y}{y \log y} = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{x}{y \log y} - \frac{1}{y} = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{x}{y \log y} = \frac{1}{y} \quad \text{--- (1)}$$

$$\begin{aligned} \text{I.F.} &= e^{\int \frac{1}{y \log y} dy} = e^{\int \frac{1/y}{\log y} dy} = e^{\ln(\log y)} \\ &= \log y \end{aligned}$$

Now multiplying eqn (1) by $\log y$ we get,

$$\log y \frac{dx}{dy} + \frac{x}{y} = \frac{\log y}{y}$$

$$\text{or, } \frac{d}{dy} (x \log y) = \frac{\log y}{y}$$

$$\begin{aligned} \text{Integrating, } x \log y &= c + \int \frac{\log y}{y} dy & \left| \begin{array}{l} \text{Let,} \\ \log y = z \\ \text{or, } \frac{dy}{y} = dz \end{array} \right. \\ &= c + \int z dz \\ &= c + \frac{z^2}{2} \end{aligned}$$

$$x \log y = c + \frac{(\log y)^2}{2}$$

which is the required solution.

Ex-2 | Solve $dx + x dy = e^{-y} \log y dy$ | x dependent

Soln : Given, $dx + x dy = e^{-y} \log y dy$

$$\text{or, } dx + x dy = e^{-y} \log y dy$$

$$\text{or, } \frac{dx}{dy} + x = e^{-y} \log y \quad \text{--- (1)}$$

$$\text{I.F} = e^{\int 1 dy} = e^y$$

Now multiplying eqn (1) by e^y we get,

$$e^y \frac{dx}{dy} + x e^y = \log y$$

$$\text{or, } \frac{d}{dy}(x e^y) = \log y$$

Now Integrating,

$$\begin{aligned} x e^y &= c + \log y \int dy + \int \frac{d}{dy} \log y \int dy \cdot dy \\ &= c + y \log y + y \end{aligned}$$

which is the required solution.

Ex-3 | Solve $(1+y^2)dx + (x - \tan^{-1}y)dy = 0$

| x dependent

Soln : ~~we~~ Given that,

$$(1+y^2)dx + (x - \tan^{-1}y)dy = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{x - \tan^{-1}y}{1+y^2} = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{1}{1+y^2} x = \frac{\tan^{-1}y}{1+y^2} \quad \text{--- (1)}$$

Now I.F = $e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1}y}$ $\left| \int \frac{dy}{1+y^2} = \tan^{-1}y \right|$

Now multiplying eqn(1) by $e^{\tan^{-1}y}$ we get,

$$e^{\tan^{-1}y} \frac{dx}{dy} + \frac{e^{\tan^{-1}y}}{1+y^2} x = e^{\tan^{-1}y} \cdot \frac{\tan^{-1}y}{1+y^2}$$

$$\text{or, } \frac{d}{dy} (x e^{\tan^{-1}y}) = e^{\tan^{-1}y} \cdot \frac{\tan^{-1}y}{1+y^2}$$

Now Integrating,

$$x e^{\tan^{-1}y} = c + \int e^{\tan^{-1}y} \cdot \frac{\tan^{-1}y}{1+y^2} dy$$

$$= c + \int z e^z dz$$

$$\left| \begin{array}{l} \text{Let,} \\ \tan^{-1}y = z \\ \text{or, } \frac{dy}{1+y^2} = dz \end{array} \right.$$

$$= c + z e^z - \int e^z dz$$

$$= c + z e^z - e^z$$

$$= c + \tan^{-1}y e^{\tan^{-1}y} - e^{\tan^{-1}y}$$

$$\text{or, } x = c e^{-\tan^{-1}y} + \tan^{-1}y - 1$$

which is the required solution.

Ex-4 | Solve $(x+2y^3) \frac{dy}{dx} = y$ | x dependent

Soln: Given that,

$$(x+2y^3) \frac{dy}{dx} = y$$

$$\text{or, } \frac{dx}{dy} = \frac{x}{y} + 2y^2$$

$$\text{or, } \frac{dx}{dy} - \frac{x}{y} = 2y^2 \quad \text{--- (1)}$$

$$\text{Now I.F} = e^{\int \frac{-dy}{y}} = e^{-\ln y} = \frac{1}{y}$$

Now multiplying eqn(1) by $\frac{1}{y}$ we get.

$$\text{or } \frac{1}{y} \frac{dx}{dy} - \frac{x}{y^2} = 2y$$

$$\text{or, } \frac{d}{dy} \left(\frac{x}{y} \right) = 2y$$

$$\text{Now Integrating, } \frac{x}{y} = \frac{2y^2}{2} + C = y^2 + C$$

$$\therefore x = y^3 + Cy$$

which is the required solution.

*** Equation reducible to linear form /

Bernoulli Equation

definition: A differential equation of the form

$\frac{dy}{dx} + py = Qy^n$ where p & Q are the function

of x or constant is called Bernoulli's differential equation.

working rule: If any differential eqn of the form $\frac{dy}{dx} + Py = Qy^n$ then dividing the eqn by y^n we get,

$$y^{-n} \frac{dy}{dx} + P \cdot y^{1-n} = Q \quad \text{--- (1)}$$

Let, $y^{1-n} = v$

$$\text{or, } (1-n) y^{-n} = \frac{dv}{dy} \cdot \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$$

So the eqn becomes,

$$\frac{1}{1-n} \cdot \frac{dv}{dx} + Pv = Q$$

$$\Rightarrow \frac{dv}{dx} + (1-n)Pv = (1-n)Q \text{ which is linear.}$$

So solve it as earlier.

Ex-1 | Solve $\frac{dy}{dx} = x^3 y^3 - xy$

Soln: Given that, $\frac{dy}{dx} = x^3 y^3 - xy$

$$\text{or, } \frac{dy}{dx} + xy = x^3 y^3 \quad \text{--- (1)}$$

Now dividing by y^3 we get,

$$y^{-3} \frac{dy}{dx} + xy^{-2} = x^3 \quad \text{--- (11)}$$

Let, $y^{-2} = v$

So that, $-2y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

So eqn (ii) becomes

$$-\frac{1}{2} \frac{dv}{dx} + xv = x^3$$

$$\text{or, } \frac{dv}{dx} - 2xv = -2x^3 \quad \text{--- (iii)}$$

which is linear.

$$\text{So, I.F.} = e^{\int -2x dx} = e^{-2 \cdot \frac{x^2}{2}} = e^{-x^2}$$

Multiplying eqn (iii) by e^{-x^2} we get,

$$e^{-x^2} \frac{dv}{dx} - 2xv \cdot e^{-x^2} = -2x^3 \cdot e^{-x^2} \quad \left| \int e^{mx} dx = \frac{e^{mx}}{m} \right|$$

$$\text{or, } \frac{d}{dx} (e^{-x^2} \cdot v) = -2x^3 \cdot e^{-x^2}$$

Now Integrating,

$$e^{-x^2} \cdot v = c + \int (-2x^3 \cdot e^{-x^2}) dx$$

$$= C - \int z e^{-z} dz$$

$$= C - \left\{ z \int e^{-z} dz - \int 1 \cdot \int e^{-z} dz \right\} \quad \left| \begin{array}{l} \text{Let, } x^2 = z \\ \therefore 2x dx = dz \end{array} \right.$$

$$= C - (-ze^{-z} + \int e^{-z} dz)$$

$$= C + ze^{-z} + e^{-z} = C + x^2 e^{-x^2} + e^{-x^2}$$

$$\therefore v = C \cdot e^{x^2} + x^2 + 1$$

$$\text{or, } y^{-2} = C \cdot e^{x^2} + x^2 + 1$$

which is the required solution.

Ex-2 | Solve $\frac{dy}{dx} + xy = xy^2$

Solⁿ: Given that,

$$\frac{dy}{dx} + xy = xy^2$$

$$\text{or, } y^{-2} \frac{dy}{dx} + xy^{-1} = x \quad \text{--- (1)}$$

$$\text{Let, } y^{-1} = v$$

$$\text{or, } -y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } y^{-2} \frac{dy}{dx} = -\frac{dv}{dx}$$

so eqn (1) become,

$$-\frac{dv}{dx} + xv = x$$

$$\text{or, } \frac{dv}{dx} - xv = -x \quad \text{--- (2) which is linear}$$

$$\text{So, I.F} = e^{\int -x dx} = e^{-\frac{x^2}{2}}$$

Multiplying eqn (2) by $e^{-\frac{x^2}{2}}$ we get,

$$e^{-\frac{x^2}{2}} \cdot \frac{dv}{dx} - xv \cdot e^{-\frac{x^2}{2}} = -x \cdot e^{-\frac{x^2}{2}}$$

$$\text{or, } \frac{d}{dx} (v e^{-\frac{x^2}{2}}) = -x \cdot e^{-\frac{x^2}{2}}$$

Now Integrating,

$$\begin{aligned} v e^{-\frac{x^2}{2}} &= c - \int x e^{-\frac{x^2}{2}} dx \\ &= c + \int e^z dz \quad \left\{ \begin{array}{l} \text{Let,} \\ -\frac{x^2}{2} = z \\ \text{or, } -x dx = dz \end{array} \right. \\ &= c + e^z \end{aligned}$$

$$(1) \quad \dots = c + e^{-\frac{x^2}{2}}$$

$$\therefore v = c \cdot e^{x^2/2} + 1$$

or, $y^{-1} = c e^{x^2/2} + 1$; which is the required solution.

Ex-3] solve $\frac{dy}{dx} + \frac{2}{x}y = \frac{y^3}{x^3}$

Soln: Given that,

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{y^3}{x^3}$$

Dividing by y^3 we get, $y^{-3} \frac{dy}{dx} + \frac{2}{x} \cdot y^{-2} = \frac{1}{x^3}$ (1)

Let, $y^{-2} = v$

$$\text{or, } -2y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } y^{-3} \frac{dy}{dx} = -\frac{1}{2} \cdot \frac{dv}{dx}$$

Now eqn (1) become,

$$-\frac{1}{2} \cdot \frac{dv}{dx} + \frac{2}{x} v = \frac{1}{x^3}$$

$$\text{or, } \frac{dv}{dx} - \frac{4}{x} v = -\frac{2}{x^3} \quad (2)$$

which is linear.

$$\text{So I.F.} = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = \frac{1}{x^4}$$

Multiplying eqn (2) by $\frac{1}{x^4}$ we get,

$$\frac{1}{x^4} \frac{dv}{dx} - \frac{4}{x^5} v = -\frac{2}{x^7}$$

$$\text{or, } \frac{d}{dx} \left(\frac{v}{x^4} \right) = -\frac{2}{x^7}$$

$$\begin{aligned} \text{Now Integrating, } \frac{v}{x^4} &= C - 2 \cdot \left(-\frac{x^{-6}}{6} \right) \\ &= C + \frac{1}{3x^6} \end{aligned}$$

$$\text{or, } v = 2x^4 + \frac{1}{3x^2}$$

$$\text{or, } y^{-2} = 2x^4 + \frac{1}{3x^2}$$

which is the required solution.

Ex-41 Solve $\frac{dy}{dx} (x^2 y^3 + xy) = 1$

Soln: Given that,

$$\frac{dy}{dx} (x^2 y^3 + xy) = 1$$

$$\text{or, } \frac{dx}{dy} = x^2 y^3 + xy$$

$$\text{or, } \frac{dx}{dy} - xy = x^2 y^3$$

Now dividing by x^2 we get,

$$x^{-2} \frac{dx}{dy} - x^{-1} y = y^3 \quad \text{--- (1)}$$

Let,

$$\text{or, } \frac{dx}{x^2} = dv$$

$$\text{or, } x^{-2} \frac{dx}{dy} = \frac{dv}{dy}$$

Now eqn (1) become,

$$\frac{dv}{dy} + vy = y^3 \quad \text{--- (2)}$$

which is linear